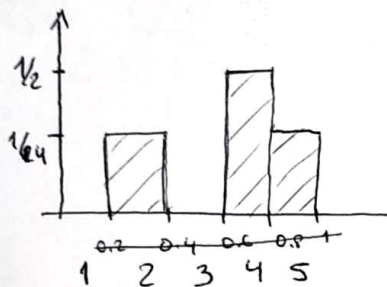
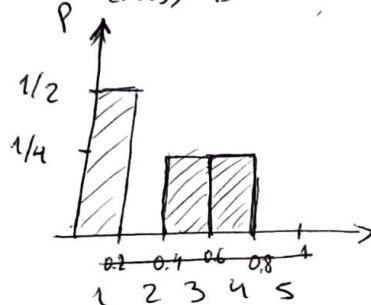


bin 1: $[0; 0.2]$ #2 (B, B)
 bin 2: $]0.2; 0.4]$ #3 (A, C, C)
 bin 3: $]0.4; 0.6]$ #3 (B, C, C)
 bin 4: $]0.6; 0.8]$ #3 (A, A, B)
 bin 5: $]0.8; 1]$ #1 (A)

class A

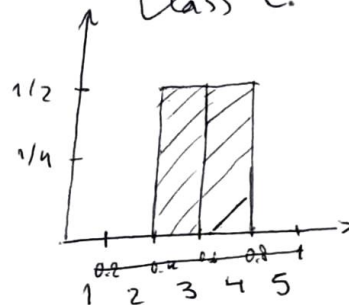


class B



$P(\text{class} = c \mid x \in \text{bin}_y)$

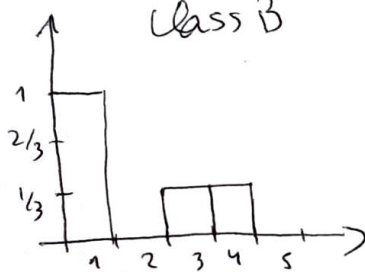
class C



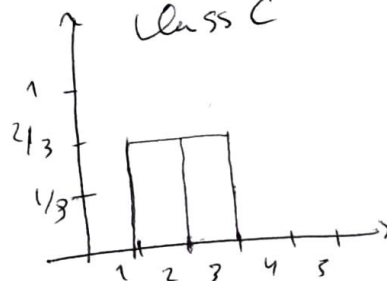
class A



class B



class C

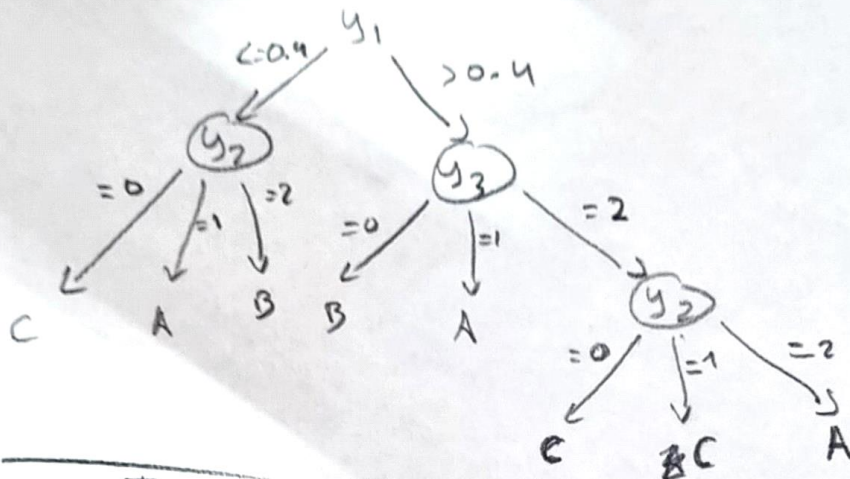


$P(x \in \text{bin}_y \mid \text{class} = c)$

$$E(\text{class} | y_3=2, y_1 \geq 0.4, y_2) = \frac{1}{4} E(\text{class} | y_2=0) + \frac{1}{4} E(\text{class} | y_2=1) + \frac{2}{4} E(\text{class} | y_2=2)$$

$$= 0$$

Logo y_2 tem ganho máximo



2)

		T			
		A	B	C	
P	A	4	1	0	5
	B	0	2	0	2
	C	0	1	4	5
		4	4	4	12

T	P
A	A
B	B
B	C
C	C
C	C
A	A
A	A
A	A
B	A
B	B
C	C
C	C

3) $F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{P + R}$

$$P_A = \frac{4}{5}$$

$$P_B = 1$$

$$P_C = \frac{4}{5}$$

$$R_A = 1$$

$$R_B = 0.5$$

$$R_C = 1$$

$$F_{1A} = 2 \times \frac{\frac{4}{5} \times 1}{\frac{4}{5} + 1} = 0.8(8)$$

$$F_{1B} = 2 \times \frac{1 \times 0.5}{1 + 0.5} = 0.6(6)$$

$$F_{1C} = 2 \times \frac{\frac{4}{5} \times 1}{\frac{4}{5} + 1} = 0.8(8)$$

A classe com menor valor F_1 é a B

$$1) E(\text{class} | y_1) \quad E(\text{class} | y_3) = \sum_i P(y_3 = z_i) E(\text{class} | y_3 = z_i)$$

$$E(p) = - \sum p_i \log_2 p_i$$

$$E(\text{class} | y_2) = \frac{3}{7} E(\text{class} | y_2 = 0) + \frac{2}{7} E(\text{class} | y_2 = 1) + \frac{2}{7} E(\text{class} | y_2 = 2)$$

$$\frac{3}{7} \left(\frac{1}{3} \log \frac{1}{3} - \frac{1}{3} \log \frac{1}{3} - \frac{1}{3} \log \frac{1}{3} \right) + \frac{2}{7} \left(0 \log 0 - \frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} \right)$$

$$\underbrace{-1 \log \frac{1}{3}}_{1.585} \quad \underbrace{1}_{1} \quad + \frac{2}{7} (0)$$

$$0.679$$

$$= 0.96$$

$$E(\text{class} | y_3) = \frac{1}{7} E(\text{class} | y_3 = 0) + \frac{2}{7} E(\text{class} | y_3 = 1) + \frac{4}{7} E(\text{class} | y_3 = 2)$$

$$\underbrace{0}_{0} \quad \underbrace{\frac{2}{7}}_{\frac{2}{7}} \quad + \frac{4}{7} \left(-\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} - 0 \log 0 \right)$$

$$\underbrace{\quad}_{\frac{6}{7}}$$

$$= 0.857$$

$$E(\text{class} | y_4) = \frac{2}{7} E(\text{class} | y_4 = 0) + \frac{3}{7} E(\text{class} | y_4 = 1) + \frac{2}{7} E(\text{class} | y_4 = 2)$$

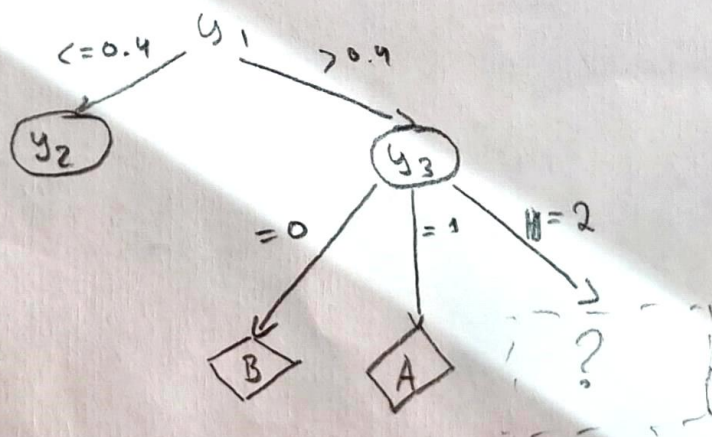
$$\underbrace{1}_{1} \quad \underbrace{\frac{3}{7} \left(-\frac{1}{3} \log \left(\frac{1}{3} \right) - \frac{2}{3} \log \left(\frac{2}{3} \right) - 0 \log 0 \right)}_{0.5283 + 0.3899} \quad + \frac{2}{7}$$

$$0.5283 + 0.3899$$

$$+ 0.393$$

$$= 0.9649$$

~~E(class)~~ logo IG e (class | y₃) e o maior



④ Y_1	Y_2	rank s $Y_1(x)$	rank s $Y_2(z)$
0,24	1	3	8
0,06	2	2	11
0,04	0	1	3,5
0,36	0	5	3,5
0,32	0	4	3,5
0,68	2	10	11
0,9	0	12	3,5
0,76	2	11	11
0,46	1	7	8
0,62	0	9	3,5
0,44	1	6	8
0,52	0	8	3,5

$$COV(X, Z) = \frac{\sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z})}{n-1} = \frac{10,5}{12-1} \approx 0,95454545$$

$$r = \frac{COV(X, Z)}{\sigma(X) \times \sigma(Z)} = \frac{0,95454545}{3,60555 \times 3,32347} = 0,07965872$$

We can conclude that there's a very weak positive correlation between the two variables. Moreover, it's so weak that it probably has no practical value.